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Investigating the effects of an LLM-based Socratic conversational agent on students' academic performance and reflective thinking in higher education ☆

Linjin Xi [a,*], Yi Zhang [a,*], Qiyun Wang [b]

[a] Faculty of Artificial Intelligence in Education, Central China Normal University, 152 Luoyu Avenue, Wuhan, China
[b] National Institute of Education, Nanyang Technological University, 50 Nanyang Avenue, Singapore, Singapore

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ABSTRACT

The rapid advancement of GenAI has created unprecedented opportunities to enhance student learning through large language models (LLMs). While LLM-based conversational agents have demonstrated potential in improving accessibility and instructional support, evidence of their impact on higher-order thinking remains limited, especially within the context of Chinese higher education. This study introduces the Socratic Intelligent Conversational Agent (S-ICA), which integrates Socratic questioning strategies with semantic generation mechanisms to promote student learning and reflective thinking in disciplinary contexts. A total of 94 university students from China participated in the study, with random assignment to an experimental group (EG) using the S-ICA and a control group (CG) using a non-Socratic conversational agent (nS-ICA). The results showed that the EG outperformed the CG in both academic achievement and reflective thinking, particularly in the dimensions of "reflection" and "critical reflection". Cognitive network analysis revealed that the EG students activated more advanced reflective pathways, linking understanding, reflection, and critical reflection processes. Although no significant differences were found in learning motivation between the groups, interviews with students indicated that the S-ICA facilitated more effective learning and deeper reflective engagement. These findings contribute to research on integrating classical pedagogical strategies into GenAI-based systems, offering insights into how such technologies can foster higher-order thinking and guide the design of future collaborative human–AI learning systems.

1. Introduction

Reflective thinking, as a critical higher-order thinking ability, has been widely recognized as fundamental for solving complex problems, fostering deep learning, and promoting innovation (Bauer, 1991; Dewey, 1922). It involves an active and sustained process through which individuals critically examine and evaluate their own experiences and beliefs, allowing cognitive restructuring and knowledge reconstruction (Schön, 1995). An increasing emphasis within higher education highlights the importance of cultivating students' ability to reflect to better prepare them for increasingly dynamic and uncertain environments (Harvey et al., 2016). Empirical evidence has shown that learners with well-developed reflective thinking skills achieve deeper conceptual understanding, greater transfer, and improved problem solving in novel contexts (Askell-Williams et al., 2007). To this end, diverse pedagogical

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* Corresponding authors.
E-mail addresses: xilinjin@mails.ccnu.edu.cn (L. Xi), zhangyi@mails.ccnu.edu.cn (Y. Zhang), qiyun.wang@nie.edu.sg (Q. Wang).

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strategies, such as dialogic classroom discourse (Cui & Teo, 2023), reflective journals or blogs (Chu et al., 2012), and collaborative group work (Lee et al., 2016)—have been employed to cultivate students' reflective awareness and critical dispositions (Cloude et al., 2021). Nevertheless, while digital technologies—and more recently advanced GenAI systems—offer promising opportunities to scale and scaffold reflective practices, their systematic integration into authentic classroom settings remains limited.

Recent advances in GenAI, particularly large language models (LLMs), offer new opportunities to address educational challenges (Yan et al., 2024; Zirar, 2023). Intelligent conversational agents (ICAs powered by LLM) can produce fluent and contextually relevant dialogue, provide immediate feedback, and deliver personalized and adaptive learning support (Kasneci et al., 2023). Empirical studies have reported encouraging results, showing that such agents can improve conceptual understanding, problem solving abilities, and knowledge transfer (Wang et al., 2025). (Liu et al., 2024). However, concerns have also emerged that excessive reliance on AI-generated answers can reduce students' critical reflection, self-regulation, and long-term engagement (Darvishi et al., 2024; Zhai et al., 2024). These concerns underscore the importance of examining not only cognitive outcomes, but also affective dimensions such as motivation and perceptions, areas that remain not adequately explored in authentic higher education contexts (Yu, 2023). Therefore, the integration of GenAI into education must address both cognitive and affective dimensions to fully understand its impact on student learning (Liu et al., 2024).

The integration of advanced AI technologies into the pedagogy requires careful alignment with robust instructional principles (Barrot, 2023; Hu et al., 2025). Researchers are increasingly focusing on the fact that learners should be guided through exploratory questioning and scaffolded instructions, rather than provided with ready-made solutions (Borchers et al., 2025). Among such approaches, the Socratic method, rooted in probing questions that encourage learners to articulate reasoning, justify claims, and confront inconsistencies, has long been recognized as effective for cultivating critical and reflective thinking (Hsu et al., 2022). Recent work has begun to explore how Socratic strategies can be implemented through agents powered by LLM. For example, Kuo et al. (2026) documented the application of the TALPer system in fifth-grade mathematics education, where it employed the Socratic dialogue method and carefully designed questioning strategies to foster reflective thinking and enhance mathematical understanding among students. However, most studies have focused on knowledge acquisition, with limited attention to how Socratic agents foster higher-order cognitive processes such as reflection and metacognitive awareness. Moreover, while Socratic dialogue was designed for one-to-one interactions, adapting it to group settings is essential for ecological validity. Although group dynamics may pose challenges like unequal participation, they also facilitate collaborative reflection and critical thinking (Fiock, 2020). Our design mitigated these risks by incorporating structured worksheets and turn-taking scaffolds to ensure active participation in small groups, thus supporting personalized and collaborative learning.

Against this backdrop, the present study introduced a Socratic Intelligent Conversational Agent (S-ICA) that integrated Socratic questioning strategies with LLM technologies to support inquiry-based learning in university classrooms. To rigorously evaluate its effectiveness, we contrasted the S-ICA with a carefully defined non-Socratic ICA (nS-ICA). Both agents operated under identical conditions in terms of tasks, time allocation, and interface; the only intended pedagogical difference was that the S-ICA employed Socratic questioning while the nS-ICA provided direct answers and clarifications without reflective prompts. We emphasize that the nS-ICA refers specifically to the control condition in this study and is not intended to be generalized to all non-Socratic agents. This design isolates the contribution of Socratic questioning while minimizing potential confounding.

The study assesses both cognitive and affective outcomes. At the cognitive level, we investigate whether the S-ICA enhances academic performance and reflective thinking. At the affective level, we explore how the S-ICA influences students' motivation and learning perceptions—critical yet underexplored dimensions in AI in education. Epistemic Network Analysis (ENA) was employed to model the co-occurrence patterns of cognitive categories in dialogue, a widely recognized method in educational research for examining learning processes and knowledge construction. By analyzing connections among reflective categories such as understanding, reflection, and critical reflection, ENA provides insights into how students' cognitive engagement evolves during collaborative tasks. In this study, we apply Epistemic Network Analysis (ENA) to reveal distinct cognitive patterns and the underlying mechanisms through which ICAs integrating Socratic prompting promote deeper reflective engagement, shedding light on how these cognitive pathways differ between the experimental and control groups. These outcomes are critical to understanding how personalized and adaptive learning support in GenAI-powered agents can promote sustained engagement and deeper learning. To this end, the following research questions were addressed:

1. Does the S-ICA approach improve learners' academic performance compared to the nS-ICA?
2. Does the S-ICA approach foster learners' reflective thinking compared to the nS-ICA?
3. Does the S-ICA approach enhance learners' learning motivation compared to the nS-ICA?
4. How do students perceive their learning experience when interacting with the two types of ICAs?

2. Literature review

2.1. Reflective thinking

Reflective thinking is defined as the capacity to actively and consciously analyze, evaluate, and integrate one's experiences, knowledge, and cognitive processes (Boyd & Fales, 1983; Paul, 2013). From different theoretical perspectives, reflective thinking has been conceptualized as comprising multiple levels and dimensions. For example, Mezirow (1998) and Kember (2000) categorized reflective thinking into four progressive stages: habitual action, understanding, reflection, and critical reflection. Among these four stages, the first two emphasize the construction of meaning from experience by learners and are considered non-reflective behaviors,

whereas the latter two involve the reevaluation of actions or ideas in response to problems or situations and are regarded as reflective behaviors. Furthermore, Mezirow (1997) defined two hierarchical levels of reflective action, in which the higher and more essential level—critical reflection—serves as the foundation for deep learning and the assessment of success by learners.

Reflective thinking has been widely recognized in higher education as a key psychological mechanism that promotes deep learning, self-directed learning, and complex problem solving (Ghanizadeh, 2017). Learners who possess reflective thinking skills are able to regulate their own learning and recognize the gap between what they know and what they need to know. Reflection plays a vital role in the learning process, helping learners improve academic achievement (Menekse, 2020), improve metacognitive abilities (Nobutoshi, 2023), and develop cross-context transfer capabilities. According to research on higher-order thinking, educational approaches that foster reflective thinking can increase student engagement and participation in flipped learning designs (Chen et al., 2019).

Among current instructional media, GenAI technologies are regarded as having new potential to support reflective learning due to their semantic interactivity and language expression capabilities. Albert and Li (2023) noted that AI instructional agents such as ChatGPT can provide immediate feedback, guide questions, and suggestions for the organization of ideas in contexts where teacher–student interaction is limited, thus promoting reflective behaviors. However, current research on AI-supported reflection has primarily focused on technological feasibility and user acceptance, with limited exploration of the underlying mechanisms by which such tools contribute to the development of higher-order thinking.

2.2. Learning outcomes of university students

Academic performance and learning motivation are critical indicators for evaluating students' learning outcomes and their sustained engagement. Academic performance reflects the depth of knowledge acquisition of students, the proficiency of their skills and the level of their problem-solving abilities (Alexandre & Enslin, 2017). Learning motivation directly influences the ability of learners to self-regulation, their investment in learning activities, and persistence over time (Schunk & Zimmerman, 2012). In university learning environments, particularly, students' ability to engage in self-directed learning and maintain long-term motivation is essential for academic success.

However, with the widespread adoption of GenAI in education, significant changes have emerged in academic performance and learning motivation. On the one hand, GenAI tools offer immediate feedback, personalized learning pathways, and adaptive resources, which have been shown to improve academic performance and motivation (Derakhshan, 2025; Gao et al., 2024). For example, ICAs have helped students clarify conceptual misunderstandings in real time, fostering deeper comprehension and mastery of knowledge (Zhai & Wibowo, 2023). On the other hand, excessive reliance on AI-generated answers can lead to overdependence, diminishing critical thinking, exploration, and potentially undermining intrinsic motivation (Han, 2026).

Moreover, understanding students' learning perceptions—their attitudes and beliefs about GenAI tools, is crucial to evaluating the effectiveness of these technologies. These perceptions are strongly influenced by the degree of personalization and adaptivity offered by the learning tool. Studies have shown that students' perceptions of AI's role significantly affect their engagement and motivation. Positive perceptions, in which students view AI as a supportive learning tool, increase motivation and foster engagement (Liu et al., 2024). In contrast, students who perceive AI as a crutch may feel less confident in their abilities, reducing motivation and engagement in the learning process. Therefore, further exploration of learning perceptions and motivation is essential, as these factors influence the sustainability of AI-supported learning and its impact on higher-order thinking skills (Yu, 2023).

2.3. Socratic questioning strategy

Socratic questioning is a pedagogical philosophy and instructional strategy centered on inquiry, emphasizing cognitive stimulation and intellectual challenge. It encourages learners to actively construct knowledge and engage in self-reflection through a process of progressive questioning (Hu, 2023). At its core, it is a dialogic method of cognitive training (Clark & Egan, 2015), designed to provoke learners to question, examine, and ultimately reconstruct their existing beliefs. The key characteristics of Socratic questioning include open inquiry, progressive exploration, and the educator's role as a facilitator of thought, elements that make it particularly effective for fostering higher-order thinking (Dalim et al., 2022).

Empirical studies have shown the efficacy of Socratic questioning in improving students' reflective awareness and critical thinking skills. For example, Hsu et al. (2022) found that structured Socratic reflection prompts significantly enhanced elementary school students' critical thinking skills. In an experimental study with secondary school students, Yang and Chung (2009) reported notable improvements in critical reasoning and self-expression after sustained Socratic dialogue sessions. Beyond the sciences, this method has gained traction in fields such as English writing instruction, teacher education, and medical ethics, where it facilitates deeper reflection on beliefs, judgments, and actions within authentic learning contexts (Dinkins & Cangelosi, 2019; Ho et al., 2023).

The pedagogical value of Socratic questioning has also been amplified through its integration with digital technologies. For example, Chang, Hung, and Yang (2024) developed an online writing platform incorporating Socratic prompts, which significantly increased students' frequency of reflection and depth of written output. In a controlled experiment, Chang, Hung, and Yang (2024) demonstrated that an AI-based dialogue system employing Socratic questioning outperformed traditional feedback mechanisms in supporting students' reflective writing. Despite its promise, the practical implementation of Socratic questioning presents challenges. Teachers must have strong questioning skills and classroom management skills to facilitate deep and flexible thinking. Simultaneously, learners may experience cognitive overload, leading to disengagement or shallow responses. As a result, optimizing the pacing of questions, personalizing prompts, and sustaining cognitive engagement through technological mediation have become critical areas for future research on integrating Socratic strategies into contemporary instructional design.

2.4. LLM-supported conversational agents

The rapid advancement of large language models (LLMs) in recent years has garnered significant attention for their application in educational contexts (Chang et al., 2024). As a hallmark of GenAI, LLMs possess advanced semantic comprehension and natural language generation capabilities, enabling their integration into intelligent educational agents that offer adaptive instructional support, automated feedback, and personalized learning pathways (Hu et al., 2024). By leveraging the dialogic capabilities of LLMs, these agents can engage in human-like interactions with learners, which are crucial to fostering personalized learning experiences (Zheng et al., 2025).

Emerging research has demonstrated the efficacy of LLM-driven educational agents in tasks such as language writing, problem solving, and critical thinking development (Liebenow et al., 2025; Razafinirina et al., 2024). However, the effectiveness of LLM-powered agents in fostering higher-order cognitive skills and strategic learning remains contested. Some studies highlight the advantages of AI-generated feedback in helping learners identify cognitive gaps and misconceptions (Naseer & Khawaja, 2025). However, concerns have been raised that excessive reliance on such agents can reduce cognitive effort and hinder self-regulation engagement, particularly in open tasks, where learners could substitute AI-generated responses for deeper cognitive processing (Zhai et al., 2024).

Although some evidence suggests that agents driven by LLM can improve cognitive engagement by providing continuous feedback and heuristic prompts (Liebenow et al., 2025), the mechanisms through which they influence multidimensional learning outcomes, such as motivation, self-regulation, and perceptions of learning, remain underexplored. This gap underscores the need for agent designs that effectively integrate LLM capabilities with dynamic pedagogical principles, balancing guidance, dialogic feedback, and reflective stimulation within a multilevel cognitive support framework. Notably, while much of the research on conversational agents has focused on traditional, non-LLM models, these agents often lack the personalized adaptability that LLMs can provide. Non-LLM systems, while valuable for task-specific feedback, are generally less flexible in tailoring interactions to learners' evolving needs, which may limit their capacity to foster long-term cognitive and affective engagement.

In response, the present study introduces the S-ICA, a hybrid design that incorporates LLM capabilities within the structured scaffolding of the Socratic method. This approach aims to enhance not only cognitive outcomes but also affective dimensions such as motivation and learning perceptions, which are critical but often underexplored in current GenAI-enhanced learning environments.

3. LLM-based S-ICA learning approach

3.1. Architecture of the S-ICA

In this study, we developed an LLM-driven S-ICA to assist students in evaluating and refining their experimental research proposals. The system architecture is depicted in Fig. 1. Instructors can provide students with learning resources and materials, such as instructional documents on experimental research design, task instructions, question templates, and sample proposals, which are integrated into the agent's knowledge database for access on demand. Students interact with the S-ICA to evaluate and optimize their original research proposals. This interaction takes place through a web browser, ensuring compatibility on both desktop and mobile devices, without the need for a dedicated mobile app. They input various components of their proposals, such as research questions, hypothesis statements and experimental methods, into the user interface. These inputs are then processed by the back-end system, where the large language model generates contextually appropriate feedback to guide students in refining their design rationale. The S-ICA employs predefined prompts and dialogue structures, with adaptive questioning based on students' input. This enables a personalized learning experience, as feedback and questions evolve according to the progress and needs of the learner. In this way, the system adapts to the characteristics of individual students, providing tailored guidance that reflects their current understanding and challenges.

User Interface. Fig. 2 illustrated the main interaction platform of the system: the Doubao web interface. Built on Doubao, a ByteDance-affiliated LLM platform, the S-ICA interface embeds Socratic dialogue and questioning strategies at its core. The system supports text input and multi-turn interaction, enabling students to engage in continuous reflection and iterative improvement during the experimental proposal evaluation task. Although the system also supports voice input, all student interactions in this study were conducted via text input through a web browser. After logging in to Doubao, students submit various components of their proposals, such as research questions, hypotheses, and experimental methods, via text input. The system, guided by predefined prompts and dialogue structures, generates real-time feedback, including counter-questions, clarification requests, and progressive inquiries. These adaptive prompts encourage students to reflect on the logical coherence and completeness of their proposals, guiding them through an iterative process of revision and refinement. Additionally, S-ICA tracks and stores the complete dialogue history, allowing students to review previous exchanges at any time, which supports ongoing revisions and reflective analysis.

Core Functional Modules. The S-ICA system was developed with a strong pedagogical orientation, integrating the Socratic method with LLM technology to construct an intelligent conversational agent with cognitive scaffolding capabilities. It incorporates four core functional modules: input comprehension, dialogue generation, dynamic feedback, and structural memory to support the ongoing reflection and knowledge construction during the evaluation of experimental research designs.

(1) input comprehension module: this module processes text input from students and performs semantic parsing, intent recognition, and a preliminary assessment of student cognitive level. For example, when a student inputs a research question, the system identifies whether it is descriptive ("What is…?") or causal ("Why/How does…?") and uses this classification to determine the stage of inquiry. Such classification provides the foundation for the generation of subsequent prompts. The design of this module follows adaptive

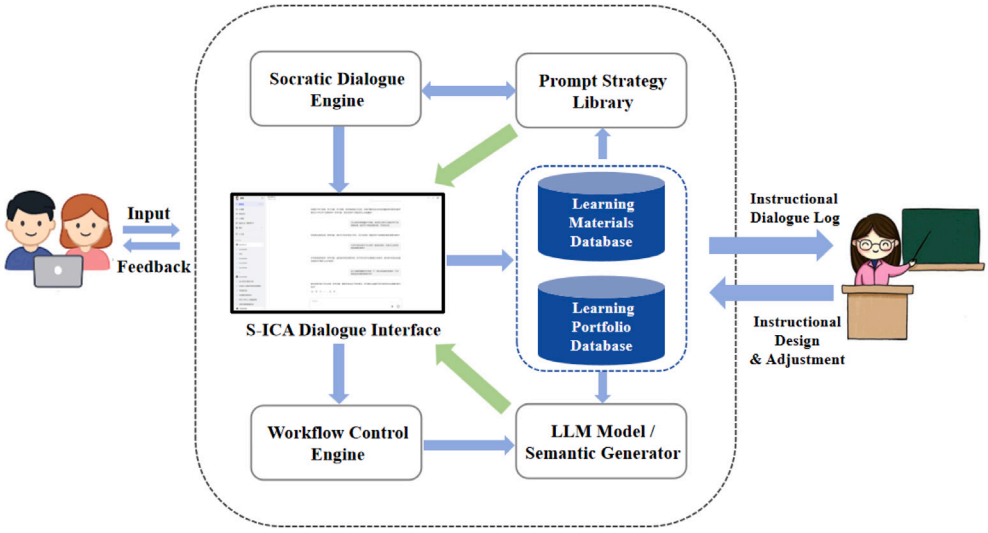


Fig. 1. System architecture of S-ICA.

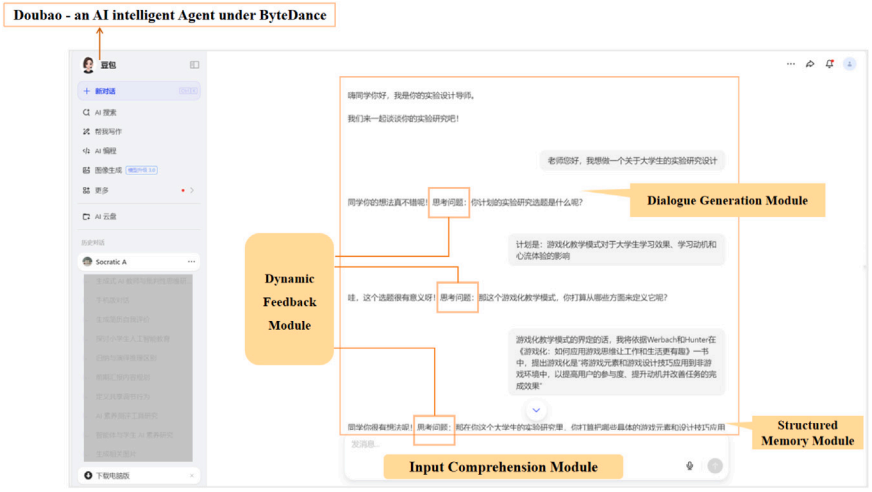


Fig. 2. Overview of the Doubao Web Interface for S-ICA.

learning principles, ensuring that the prompts are matched to the current level of reasoning of the students, thus preventing both oversimplification and cognitive overload.

(2) dialogue generation module: this module generates reflective and context-specific questions embedded within a structured prompting framework. Drawing on Socratic principles (Paul & Elder, 2008), it employs techniques such as counterquestions, clarification, and progressive inquiry to elicit deeper reasoning. For example, if a student proposes *"The impact of gamified teaching models on primary school students' academic achievement"*, the system may respond: *"How will you operationalize 'academic achievement' in measurable terms?"* or *"What mechanism links gamification to learning outcomes?"*. By advancing learners from broad ideas to more rigorous formulations, this module scaffolds critical reflection and methodological precision.

(3) dynamic feedback module: this module adapts the depth and specificity of system responses based on the complexity and precision of the student's input. A novice level statement such as *"The variable is classroom participation"* triggers concrete guidance *(e.g., "How will participation be measured or observed in your study?")*. In contrast, when a student provides a more elaborate proposal, the system escalates the investigation by prompting consideration of alternative explanations or methodological limitations. This ensures differentiated scaffolding that matches the developmental stages of the students, fostering both engagement and higher-order thinking.

(4) structural memory module: this module maintains dialogue history and semantic anchors across interactions. For example, if a student initially defines "academic achievement" as test performance, the system recalls this definition in subsequent turns and prompts: *"Does your method of data collection remain consistent with your earlier definition of achievement?"*. By referencing prior input,

the system maintains contextual coherence and promotes iterative refinement. This design supports reflective practice by ensuring continuity and alignment throughout the inquiry process.

Prompt Engineering Design. The prompting mechanism operationalizes the dialogue generation and dynamic feedback modules through a workflow-based design. Each workflow node corresponds to a component of the research design task (e.g., research questions, hypotheses, variables, methods, etc.). When learner input contains specific trigger keywords (e.g., "hypothesis", "sample size", "data collection"), the system automatically activates the corresponding workflow and retrieves a tailored prompt template. For example, in the hypothesis formulation stage, an input containing causal terms such as "effect" or "impact" triggers prompts like: *"What mechanism links your independent and dependent variables?"*. Similarly, in the methods stage, references to "sample size" activate prompts such as: *"How will you justify the adequacy of your chosen sample for detecting significant effects?"*. These templates are pre-configured but dynamically adapted in real time to align with the student's context and previous dialogue history.

The design was iteratively refined through pilot testing. Students interacted with prototype versions of the system and provided feedback on prompt clarity, difficulty, and relevance. Their feedback led to two key revisions: (1) broadening the knowledge base of the system and integrating a controlled web search interface to expand the scope of responses, and (2) reorganizing the prompt library into a matrix structure. The tiered structure progresses from clarification to justification (e.g., *"Why is gamification an appropriate intervention for this population?"*), and finally to refinement (e.g., "How will you control for differences in prior knowledge?").

In addition to workflow rules, each prompt template encodes explicit skills and constraints. For example, skills include guiding the student to articulate their reasoning, initiating clarification before advancing the dialogue, and encouraging reflection rather than passive acceptance. Constraints ensure pedagogical integrity: all prompts must follow the *"Thinking question:"* format, and the system is prohibited from directly providing answers to students. These specifications guarantee that the interaction remains inquiry-driven, aligns with Socratic principles, and fosters adaptive learning through structured, yet open-ended questioning.

3.2. S-ICA learning approach

3.2.1. Application context

The instructional practice in this study was designed as an extracurricular self-directed learning activity that extended the formal curriculum on research design theory and methodology. After completing foundational coursework, second-year undergraduates majoring in Educational Technology engaged in a six-week task to collaboratively construct a comprehensive experimental research proposal. The task was structured into weekly stages, each targeting a specific subtask: (1) clarifying the research question, (2) formulating hypotheses, (3) identifying variables, (4) designing implementation procedures and (5) evaluating the overall design. The students were organized into teams of five to six, each member responsible for contributing to the group proposal. To ensure equity between conditions, all students followed identical task instructions, used the same learning materials, and accessed the agents through a common institutional portal and browser-based interface.

Agent conditions and interaction characteristics. To evaluate the impact of ICAs on academic performance, reflective thinking and learning motivation, students were randomly assigned to one of two conditions: an experimental group (EG) and a control group (CG). Both groups received identical task requirements, time allocations, materials, and user interface. The sole intended pedagogical difference concern the agent's feedback strategy.

- **S-ICA (EG).** The S-ICA implemented a Socratic questioning strategy: it prioritized clarification, justification, and refinement through reciprocal prompts rather than supplying immediate answers. Typical turns included brief counter-questions (e.g., 'What assumptions underlie this hypothesis', 'How would you operationalize this variable in measurable terms') that pressed students to articulate reasoning, examine coherence, and iteratively improve design decisions.
- **nS-ICA (CG).** The nS-ICA provided direct, task-aligned feedback and procedural clarifications without reflective or recursive prompts. Its turns focused on definitional guidance, checklists, and examples relevant to the submitted content (e.g. 'This variable can be measured via standardized test scores' 'A feasible data collection procedure is a classroom survey with informed consent'), without follow-up probing, metacognitive cues, or dialogic escalation. In other words, the nS-ICA could provide practical cognitive, behavioral, or procedural guidance, but it deliberately did not engage in Socratic inquiry.

Model platforms and parity controls. The two agents were deployed with different LLM backends (Doubao for the S-ICA and ERNIE for the nS-ICA), due to institutional access constraints and token-based interface compatibility, rather than deliberate pedagogical differentiation. Despite this back-end divergence, both agents were delivered through an identical COZE-based web platform interface, ensuring uniformity in access mode, operational procedures, and user experience. Parity controls were also applied by aligning the task design, learning materials, and usage protocols between conditions. Consequently, the only intended experimental manipulation concerns the feedback strategy, thus strengthening the attribution of any observed group differences to the presence or absence of Socratic questioning, rather than to disparities in interface or system access.

Prior to the main study, the S-ICA was pilot-tested in two stages. First, five master's and doctoral-level students in Educational Technology reviewed the system to evaluate the clarity and consistency of Socratic prompts, leading to iterative refinements of prompt wording and sequencing. Second, one undergraduate team engaged in a pre-class trial session, during which real-time feedback was collected and used to further optimize the system's usability and alignment with classroom dynamics. This multi-stage pilot process ensured that the implemented version of the S-ICA was both pedagogically valid and technically robust.

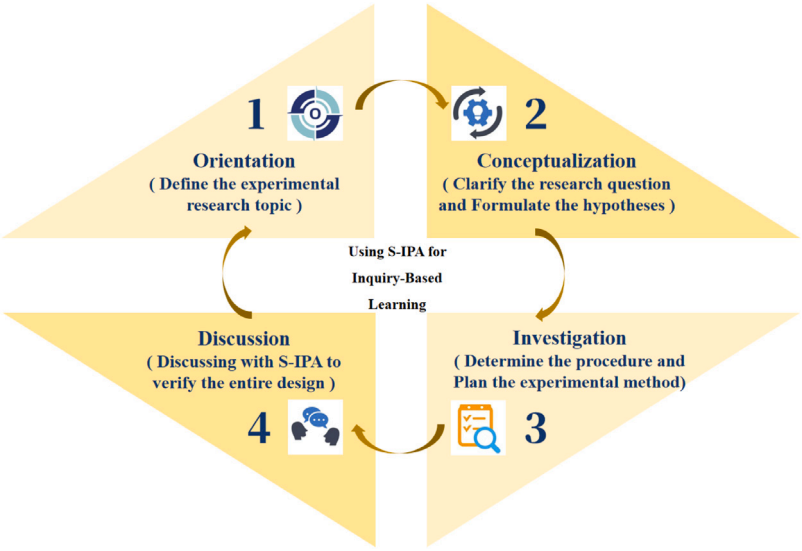


Fig. 3. Inquiry-based learning framework Using S-ICA.

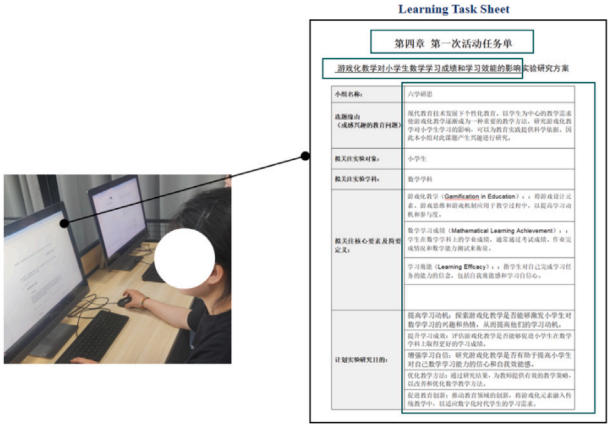


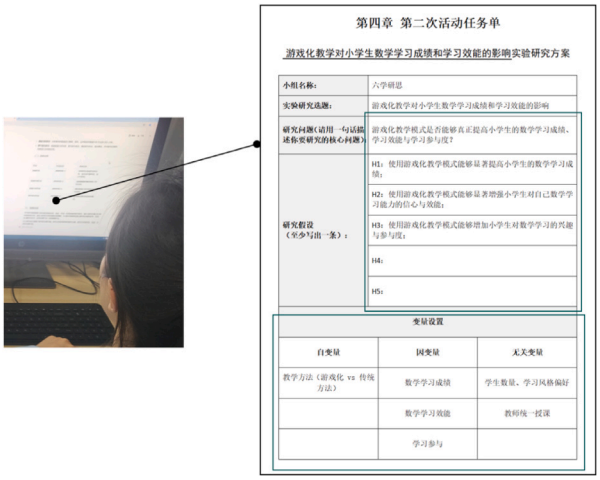
Fig. 4. Complete Task 1 of the orientation phase.

3.2.2. Learning activities

Pedaste et al. (2015) propose an inquiry-based learning (IBL) framework comprising five general phases: Orientation, Conceptualization, Investigation, Conclusion, and Discussion. This cycle provides learners with a structured and scaffolded approach to inquiry activities and has been shown to support the development of complex inquiry-driven learning processes, as well as inform the design of instructional guidance that optimizes learning efficiency. Following the IBL framework, students begin the inquiry process with Orientation, during which they not only familiarize themselves with the topic to be explored but also gain a deeper understanding of the problem that needs to be addressed. In this study, a scaffolded inquiry-based learning model was adopted for the self-practice phase. The experimental group used the S-ICA to facilitate each inquiry stage, enabling students to iteratively refine and revise their experimental proposals. Given that the students did not have enough time to conduct actual experiments, collect data, and perform data analysis, the original "Conclusion and Data Interpretation" phase was omitted. Instead, the student teams completed only the design of their experimental research proposals. The learning model, grounded in the IBL framework and standard research design steps, is illustrated in Fig. 3.

(1) Orientation phase During orientation, the students' primary task was to define their experimental research topic. Initially, they identified an educational phenomenon or behavioral issue of interest and tentatively delineated the scope and population of their experiment. The students then progressively narrowed down a broad problem into a specific and well-defined research topic. This focusing process helped establish a clear starting point for research. The detailed workflow for this phase is shown in Fig. 4.

(2) Conceptualization phase In conceptualization, the students clarified their research question, constructed testable hypotheses, and identified key variables for their chosen topic. This stage began with an in-depth analysis of the research objective, gradually translating it into operational research questions. Based on this foundation, students were expected to propose hypotheses supported



第四章 第二次活动任务单

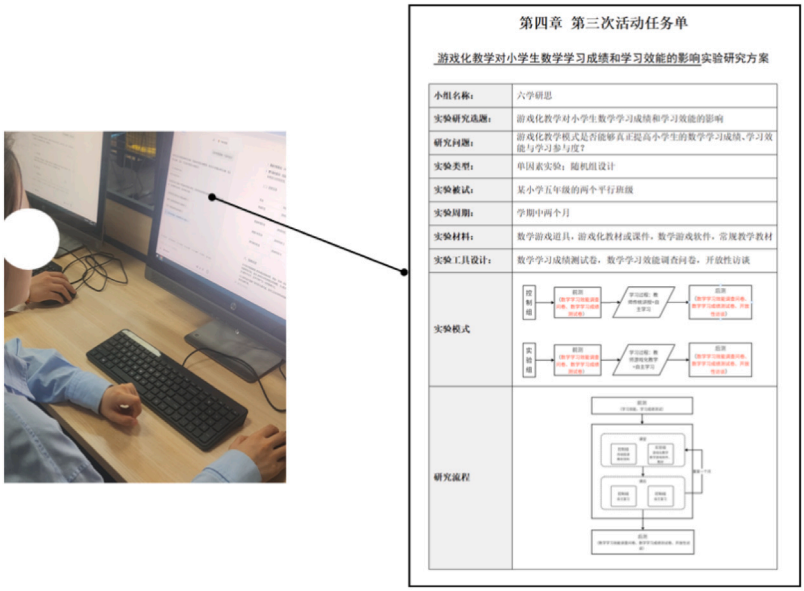
游戏化教学对小学生数学学习成绩和学习效能的影响实验研究方案

小组名称：	六学研思
实验研究选题：	游戏化教学对小学生数学学习成绩和学习效能的影响
研究问题（请用一句话描述你要研究的核心问题）	游戏化教学模式是否能够真正提高小学生的数学学习成绩、学习效能与学习参与度？
研究假设（至少写出一条）：	H1：使用游戏化教学模式能够显著提高小学生的数学学习成绩；
	H2：使用游戏化教学模式能够显著增强小学生对自己数学学习能力的信心与效能；
	H3：使用游戏化教学模式能够增加小学生对数学学习的兴趣与参与度；
	H4：
	H5：

变量设置

自变量	因变量	无关变量
教学方法（游戏化 vs 传统方法）	数学学习成绩	学生数量、学习风格偏好
	数学学习效能	教师统一授课
	学习参与	

Fig. 5. Complete Task 2 of the conceptualization phase.



第四章 第三次活动任务单

游戏化教学对小学生数学学习成绩和学习效能的影响实验研究方案

小组名称：	六学研思
实验研究选题：	游戏化教学对小学生数学学习成绩和学习效能的影响
研究问题：	游戏化教学模式是否能够真正提高小学生的数学学习成绩、学习效能与学习参与度？
实验类型：	单因素实验；随机组设计
实验被试：	某小学五年级的两个平行班级
实验周期：	学期中两个月
实验材料：	数学游戏道具，游戏化教材或课件，数学游戏软件，常规教学教材
实验工具设计：	数学学习成绩测试卷，数学学习效能调查问卷，开放性访谈
实验模式	控制组 前测 → 学习过程：教师传统讲授一主学习 → 后测 实验组 前测 → 学习过程：教师游戏化教学一助主学习 → 后测
研究流程	前测 → 课 → 后测

Fig. 6. Complete Task 3 of the investigation phase.

by relevant literature and explicitly define independent, dependent, and control variables. Thus, conceptualization provided the theoretical and logical foundations for the subsequent design and implementation of research. The specific steps for this phase are shown in Fig. 5.

(3) Investigation phase

During investigation, the students developed a concrete experimental design based on their previously articulated research questions and hypotheses. They specified procedural details, including experimental protocols and workflows, with an emphasis on operational feasibility. Detailed sub-steps for this phase appear in Fig. 6.

(4) Discussion phase

In the discussion phase, students engaged in a deep dialogue with the S-ICA to evaluate and refine their experimental research proposals. Through iterative questions about their chosen topic, variable operationalization, hypothesis validity, and experimental procedures, the agent guided the students to uncover logical gaps and identify opportunities for improvement. This stage prioritized the cultivation of critical and reflective thinking, encouraging students to self-correct and enhance their proposals based on agent prompts. The key distinction between the EG and CG during this phase was the type of conversational tool used: the EG interacted with an agent that embedded Socratic questioning strategies to foster deeper reflection, whereas the CG used a generic conversational agent without specialized research-design guidance. Fig. 7 highlights this contrast in agent functionality and pedagogical support.

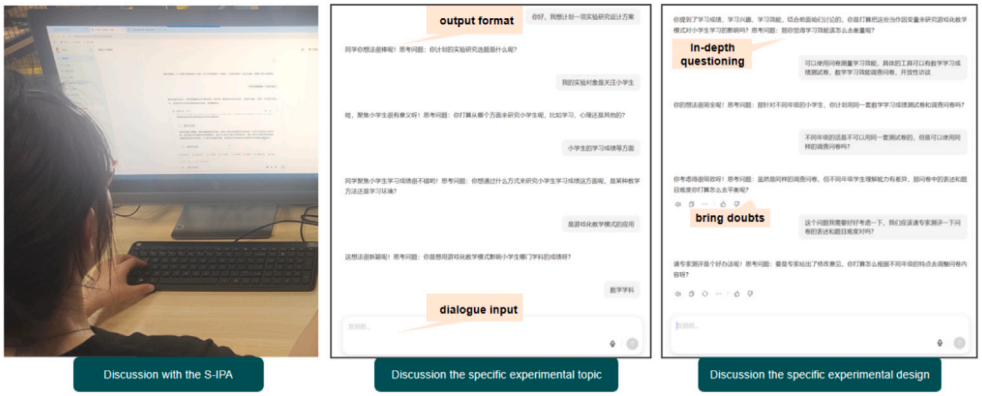


Fig. 7. Using S-ICA approach to complete task 4 of the discussion phase and some dialogue.

4. Method

4.1. Research background and participants

This study was implemented during the fall semester of 2024 in a university of teachers in central China. Participants were second-year undergraduates majoring in Educational Technology who were enrolled in the required course *Research Methods in Educational Technology*. The course lasted 16 weeks and consisted of weekly lecture sessions and practical training sessions. The instruction covered a variety of topics, including research design, literature review, survey and experimental methodologies, and data analysis, ensuring that the students had a sufficient foundation to participate in the experimental task designed for this study. A total of 94 students participated. Forty-eight students (28 female; mean age = 19.00 ± 0.22 years) were assigned to the experimental group, the remaining 46 students (25 female; mean age = 19.00 ± 0.15 years) formed the control group. In both conditions, the agents were accessed through the same institutional portal and browser interface to minimize extraneous differences; the intended pedagogical contrast only concerned the feedback strategy, i.e., Socratic questioning versus direct clarifications.

All participants were randomly assigned to 16 subgroups of five to six students each. Within these subgroups, students collaborated to design and refine an experimental research proposal over a six-week intervention period. The use of group-based tutoring reflected authentic classroom practices in the course and enabled manageable implementation across a relatively large cohort. However, group work necessarily involves the potential interdependence of data between students within the same subgroup. To address this issue, statistical analyzes explicitly incorporated group membership as a clustering factor and employed models that account for nested data structures, thereby mitigating risks of inflated significance due to non-independence.

The intervention period lasted six weeks, during which the students iteratively developed their research proposals in weekly stages. Although relatively brief in duration, the design was intentionally aligned with the temporal constraints of the semester schedule and the scope of the course project. As such, the findings are interpreted as evidence of short-term impacts within a defined academic context, and the limitations of long-term transferability and generalizability are acknowledged in the discussion. Throughout the study, both groups were instructed by the same professor, who had over 20 years of teaching experience, and followed an identical syllabus to ensure the equivalence of the instructional context. All participants had prior experience using GenAI tools. Before starting the study, ethical approval was obtained from the university's Institutional Review Board.

4.2. Experimental procedure

Fig. 8 illustrates the overall experimental procedure. Oriented to theoretical instruction and proposal design for experimental research methods, we implemented a quasi-experimental intervention lasting six weeks. Each week consisted of two 45 min theory sessions and two 45 min practical sessions. The intervention was divided into three phases: a pre-test phase, an intervention phase using ICAs, and a post-test phase.

Phase 1 (Pretest): All 94 students completed the pre-test measures, which included questionnaires that assessed reflective thinking and learning motivation, as well as a knowledge pre-test on experimental research methods (30 min). Independent samples t-tests confirmed that, before the test, there were no significant differences between the experimental and control groups in reflective thinking ($t = 0.45$, $p = 0.65$), learning motivation ($t = 0.30$, $p = 0.76$), or knowledge test scores ($t = 0.26$, $p = 0.795 > 0.01$).

Phase 2 (Intervention): Both groups participated in weekly theoretical lectures and self-practice sessions. During theory lectures (90 min per week), students in both groups attended face-to-face instruction in a smart classroom, where they were taught the fundamental principles, core concepts, and typical methodologies of experimental research. In each self-practice session, the students were required to use the assigned ICAs to evaluate and progressively refine an experimental research design proposal.

Before starting the task, all participants received a 15 min training session that provided unified instruction on system operation, task template structure, and the protocol to initiate interaction. Specifically, every student was instructed to start by entering the

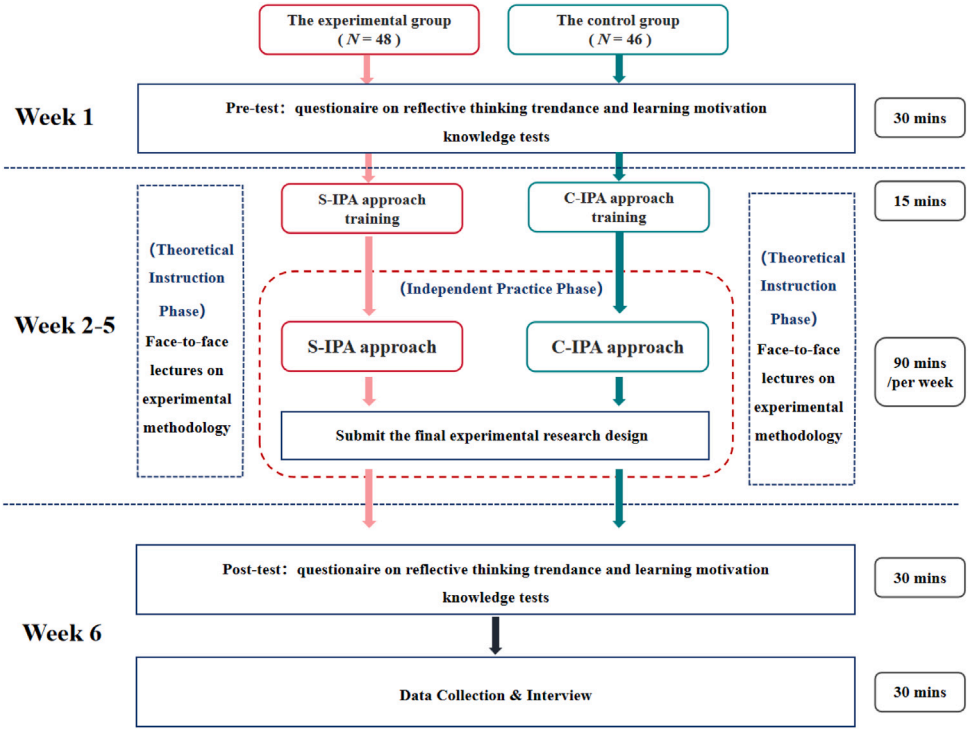


Fig. 8. The experimental procedure.

standardized initiating prompt: *I plan to conduct an experimental study. The topic is…*. This ensured that any subsequent differences in interaction were arose solely from the agents' response mechanisms rather than variations in task initiation or user familiarity.

Both agents were accessed through the same coze-based web interface to ensure parity of user experience, but they were instantiated in different back-end models. The S-ICA (experimental group), deployed in Doubao, embedded a structured prompt-engineering framework that parsed learner input and generated Socratic-style cognitively scaffolded follow-up questions to stimulate reflection and reasoning. In contrast, nS-ICA (control group), instantiated in ERNIE but delivered through the identical interface, provided direct and surface-level responses without recursive probing or structured pedagogical scaffolding. Thus, the only experimental manipulation was related to the feedback strategy, isolating the effect of Socratic questioning on learning outcomes.

Phase 3 (Posttest): At the end of the intervention, all participants completed a post-test (30 min) comprising the same reflective thinking and motivation for the learning questionnaires, as well as a knowledge test, if applicable. In addition, each subgroup's experimental research proposal was coded and rated according to a standardized rubric. We also collected dialogue logs between learners and their respective agents and conducted semi-structured interviews with selected students to gain deeper insight into their learning experiences.

4.3. Measurement instruments

The instruments used in this study comprised pre- and post-test knowledge assessments, a learning motivation questionnaire, and a reflective thinking questionnaire. In addition to these standardized measures, all dialogue data were derived exclusively from text-based interactions, thereby eliminating variability due to input modality, and were subsequently coded for cognitive network analysis. Semi-structured interviews were also conducted to provide qualitative insights into participants' learning experiences. Finally, the academic performance of the students was evaluated through the quality of their completed experimental research proposals.

4.3.1. Academic performance

Within the inquiry-based learning process, a student's mastery of conceptual knowledge reflects both the depth of their understanding and the development of research competencies. Consequently, conceptual knowledge scores were chosen as a key indicator of student academic performance. The pretest and posttest instruments were designed around the core content of experimental research methods in educational technology, covering essential concepts such as the logic of experimental design, variable types and operational definitions, random assignment procedures, validity and reliability issues in experimental research, and basic statistical analysis techniques. Both tests were co-developed by two experts with extensive experience in educational

technology research and teaching. They were then reviewed for content validity by one full professor and one associate professor; their evaluation confirmed that the tests demonstrated strong validity.

In addition, the quality of the experimental research proposal of each group served as an indicator of academic performance and research capability; it was therefore considered a critical student outcome. To evaluate the quality of the proposal, this study adapted the research proposal scoring rubric developed by Zhang et al. (2019), which has been widely used in the instruction in educational technology and research methods to assess students' design skills and the quality of academic writing. In line with the context of the course, the rubric was modified to encompass six core dimensions (see Appendix 7): (1) Research topic, (2) Research question, (3) Theoretical rationale, (4) Research content, (5) Research methodology and (6) Research procedures. Each dimension was rated using a five-level scale ranging from 2 to 10 points, with higher scores indicating better performance. Specifically, ratings were assigned in increments of two points (2 = poor, 4 = needs improvement, 6 = satisfactory, 8 = good and 10 = excellent). Higher scores indicate higher quality. The total possible score for each proposal was 60 points. The rubric was reviewed by the course instructor and an additional expert in educational technology, both of whom confirmed its appropriateness for evaluating the quality of students' research designs. To ensure objectivity, two teaching assistants independently scored each proposal. Inter-rater reliability, assessed by Cohen's kappa ($\kappa = 0.82$), indicated substantial agreement between raters (Cohen, 1988; Landis & Koch, 1977).

4.3.2. Reflective thinking

To assess the reflective thinking ability of the students, we employed a scale adapted from Kember et al. (1999), which measures four dimensions: Habitual Action, Understanding, Reflection, and Critical Reflection. Each dimension comprises four elements, for a total of 16 elements. To suit the context and population of this study, the wording of the items was revised as needed. The validity of the content was reviewed by two experts in educational technology, who confirmed the suitability of the scale to evaluate reflective thinking in the context of experimental research design. For example, one element reads: 'When I encounter a solution that appears correct, I reflect on my original viewpoint'. Each item uses a five-point Likert scale (1 = strongly disagree, 5 = strongly agree). In this study, the Cronbach's α values for the four subscales, Habitual Action, Understanding, Reflection, and Critical Reflection, were 0.82, 0.88, 0.86, and 0.89, respectively. The overall Cronbach's α for the reflective thinking scale was 0.86, indicating strong internal consistency.

4.3.3. Learning motivation

To measure students learning motivation, we use the instrument developed by Chen et al. (2015), which is a revision of the Motivated Strategies for Learning Questionnaire (MSLQ; Pintrich et al., 1993). The scale consists of six items on two subscales: Intrinsic Motivation (three items; e.g., "I enjoy learning challenging knowledge and materials because it allows me to discover new things") and Extrinsic Motivation (three items; e.g., "Achieving good grades in experimental research methods is a major source of satisfaction for me"). All items were rated on a five-point Likert scale (1 = strongly disagree, 5 = strongly agree), with higher scores reflecting stronger motivation. In the present study, the scale demonstrated acceptable internal consistency, with Cronbach's $\alpha = 0.81$. The validity of the content was reviewed by two domain experts, who confirmed its suitability for use in the context of the instruction of experimental research methods.

4.3.4. Interview questions

In addition, a semi-structured interview protocol was developed to capture students' experiences and perceptions of using the intelligent conversational agent. The interview questions were adapted from Hwang et al. (2009) and included five specific questions. The specific interview outline is provided below. Sixteen students — eight from the experimental group (labeled S1–S8) and eight from the control group (labeled C1–C8) — were randomly selected for individual interviews lasting approximately 20 min each. All interviews were recorded and transcribed verbatim. Two experienced researchers then performed qualitative content analysis. In instances of coding disagreement, the researchers reviewed and discussed the coding until reaching consensus.

(1) In what ways does the S-ICA/nS-ICA learning approach differ from other learning methods you have used before?

(2) Overall, what are the main advantages of the S-ICA/nS-ICA learning approach?

(3) What are the greatest benefits you personally gained from using the S-ICA/nS-ICA approach? What did you learn most from it? Please be specific.

(4) What aspects of the S-ICA/nS-ICA learning approach do you think need improvement? Please specify.

(5) Would you recommend the S-ICA/nS-ICA learning approach to your classmates or friends? Why or why not?

4.4. Data collection and analysis

All quantitative analyses were conducted using a combination of *Jamovi 2.3.24* and *R 4.3.2* (with the `lme4`, `car`, and `coin` packages). Descriptive statistics were first computed for all outcome measures. Normality was examined using the Kolmogorov–Smirnov and Shapiro–Wilk tests, and homogeneity of variance was assessed with Levene's test.

For the conceptual knowledge test, repeated measures were analyzed using a two-level linear mixed effects model (LMM), with students (Level 1) nested within collaborative groups (Level 2). Time (pre vs. post) and Condition (S-ICA vs. nS-ICA) were modeled as fixed effects and pretest scores were included as covariates where appropriate. Welch's t test was used as a robustness check against variance heterogeneity, and ANCOVA models with HC3 robust standard errors were estimated for supplementary confirmation of group differences. For the research proposal quality, where each group submitted a single collective product ($n = 8$ per condition), multiple robust analyses were employed due to the small sample size: Welch's t test, Mann–Whitney U test, and permutation tests

Table 1
Comparison of the existing ship path planning methods.

Code category	Abbreviation	Definition	Example
Habitual Action	HA	Routine statements about the task or context that focus solely on completion without deeper reflection.	"I followed the instructor's steps exactly as required and did not think further".
Understanding	U	Restating, explaining, or paraphrasing a concept or information, indicating initial grasp of the content.	"According to the literature, an experimental design must include random assignment and a control group to ensure fairness".
Reflection	R	Questioning or re-examining a previous viewpoint, hypothesis, or method to explore its rationale and limitations.	"I'm wondering why we selected this particular variable—does it truly relate to the research question?"
Critical Reflection	CR	Evaluating underlying assumptions, premises, or contextual factors, or proposing alternative interpretations, reflecting higher-order critical thinking.	"If we don't adequately control for confounding variables, the results could be influenced by external factors—does this hypothesis remain rigorous?"

with 10,000 iterations. Standardized effect sizes (Cohen's d, Hedges' g, and common-language effect size) were reported with 95% confidence intervals.

For the reflective thinking questionnaire, ANCOVA was used to compare post-test scores across conditions while controlling for pre-test scores. Subscale analyses were performed for Habitual Action, Understanding, Reflection, and Critical Reflection. Because these were self-reported measures, the results were interpreted as the perceived levels of reflective engagement of the participants. For the learning motivation questionnaire, ANCOVA was also applied with pretest motivation as a covariate.

To capture deeper cognitive processes, all dialogue transcripts from student-agent interactions were coded for reflective thinking categories (see Table 1) (Habitual Action, Understanding, Reflection, and Critical Reflection) following the framework of Kember et al. (2000). Two trained raters independently coded all dialogues. The interrater reliability reached $\kappa = 0.82$ and ICC = 0.85, indicating strong agreement. Discrepancies were discussed and consensus was reached after three rounds of deliberation.

The coded data were used for Epistemic Network Analysis (ENA) to model the co-occurrence patterns of reflective categories. ENA is a method for visualizing and analyzing relationships between categories in dialogue. The analysis was performed using the online ENA tool (https://app.epistemicnetwork.org) to construct and visualize epistemic networks. The data were organized into a matrix with dialogues (or group interactions) as rows and reflective thinking categories as columns. This matrix was input into ENA for network construction. ENA modeled the co-occurrence patterns by calculating the frequency of category pairings, visualizing the networks as graphs where nodes represented categories and edges represented co-occurrences. Group-level contrasts were tested by comparing ENA dimensions (e.g., centrality, clustering, and density) using independent-sample t-tests to identify significant differences in network structures between the experimental (S-ICA) and control (nS-ICA) groups. Key parameters included a sliding window of 5 utterances, Z-score normalization, PCA for optimal separation of categories, and a threshold of 0.1 to filter weak connections, ensuring focus on significant co-occurrences.

Finally, semi-structured interview data were analyzed using thematic analysis in *ATLAS.ti 25*. Two researchers independently performed open coding, axial coding, and theme refinement, following the procedures of Zhang and Creswell (2013). Disagreements were resolved through discussion until full agreement was reached, ensuring the reliability and transparency of the qualitative findings.

5. Results

5.1. Results on RQ1

5.1.1. Conceptual knowledge assessment

Table 2 reports the descriptive statistics for the conceptual knowledge test. Both groups demonstrated substantial pre-to-post gains. The S-ICA group (EG, $n = 48$) improved from $M = 68.05$ (SD=1.19) to $M = 86.44$ (SD=1.04), while the nS-ICA group (CG, $n = 46$) rose from $M = 67.60$ (SD=1.68) to $M = 85.17$ (SD=1.90). To assess group differences, we fitted a two-level linear mixed model with students (Level 1) nested within groups (Level 2), including pre-test scores as a covariate. The results showed a robust main effect of Time ($\hat{\beta} = 17.568$, $SE = 0.307$, $z = 57.23$, $p < 0.001$), confirming significant overall improvement. The main effect of Condition was not significant ($\hat{\beta} = 0.450$, $SE = 0.304$, $z = 1.48$, $p = 0.138$). The Time × Condition interaction was marginal ($\hat{\beta} = 0.821$, $SE = 0.430$, $z = 1.91$, $p = 0.056$, 95% CI [–0.03, 1.67]), suggesting that the S-ICA group achieved a slightly larger gain. The effect size was small to moderate (Hedges' $g = 0.35$).

As robustness checks, we compared gain scores with Welch's t test, which again showed no significant differences ($t(70.31) = 1.70$, $p = 0.093$), and conducted an ANCOVA with post-test as the outcome, pretest as covariate, and Condition as fixed factor, using HC3 robust errors. The ANCOVA yielded a significant Condition effect ($\hat{\gamma} = 1.366$, $SE = 0.329$, $t = 4.15$, $p < 0.001$), reflecting an adjusted

Table 2
Descriptive and inferential statistics for the conceptual knowledge test.

Group/Test	N	Pre-test M (SD)	Post-test M (SD)	Estimate	SE	t/z	p-value
EG (S-ICA)	48	68.05 (1.19)	86.44 (1.04)	–	–	–	–
CG (nS-ICA)	46	67.60 (1.68)	85.17 (1.90)	–	–	–	–
LMM: Time (main effect)	–	–	–	17.568	0.307	57.23	< 0.001
LMM: Condition (main effect)	–	–	–	0.450	0.304	1.48	0.138
LMM: Time × Condition	–	–	–	0.821	0.430	1.91	0.056
ANCOVA: Condition	–	–	–	1.366	0.329	4.15	< 0.001

Table 3
Research proposal quality: descriptive statistics and between-group effect size (team-level, $n = 8$ per condition).

Measure	S-ICA (EG)	nS-ICA (CG)	Notes
N (teams)	8	8	One proposal per team
Mean (proposal score)	48.09	40.43	Higher is better
SD	4.65	3.90	–
Between-group effect (S-ICA minus nS-ICA)			
Mean difference	7.66		Unadjusted difference in means
Hedges' g	1.69		Small-sample corrected Cohen's d
95% CI for g	[0.57, 2.80]		Normal-approximation CI
CLES	0.88		Probability S-ICA > nS-ICA

mean difference of about one point (S-ICA: 86.49; nS-ICA: 85.12). Given the sensitivity of this test to variance heterogeneity and its disregard for within-student dependence, this finding is interpreted as supplementary to the mixed-model results.

Taken together, these results indicate that both groups made substantial gains in conceptual knowledge. The S-ICA condition showed at most a modest, marginal advantage over the nS-ICA condition, suggesting that Socratic scaffolding may provide incremental benefits, although the effect remains small and warrants careful interpretation.

5.1.2. Experimental research proposal quality

The quality of the experimental research proposals was evaluated at the group level, since each team of five to six students collaboratively submitted a single final design. In total, sixteen proposals were collected ($n_{EG} = 8$, $n_{CG} = 8$). Each proposal was independently scored by two experienced raters in educational technology, and interrater reliability was substantial (Cohen's $\kappa = 0.713$ for EG and 0.670 for CG; Landis & Koch, 1977).

Descriptive statistics indicated that the experimental groups (S-ICA) obtained higher scores ($M = 48.09$, $SD = 4.65$) than the control groups (nS-ICA; $M = 40.43$, $SD = 3.90$), yielding an average difference of 7.66 points (see Table 3). Given the limited number of independent observations, statistical inference was performed using robust procedures. The Mann–Whitney U test revealed a significant difference ($U = 60.0$, exact $p = 0.002$). A nonparametric Mann–Whitney test converged on the same conclusion ($U = 60.0$, exact $p = 0.002$), and a permutation test with 10,000 random reallocations of group labels further confirmed the robustness of this effect ($p = 0.001$).

The standardized mean difference was large: Cohen's $d = 1.79$ (Hedges' $g = 1.69$, 95% CI [0.57, 2.80]). The corresponding common language effect size (CLES) was approximately 0.88, indicating an 88% probability that a randomly selected S-ICA team would achieve a higher proposal score than a randomly selected nS-ICA team.

Taken together, these convergent analyses demonstrate that the proposals produced by students interacting with the S-ICA were of substantially higher quality than those developed with the nS-ICA, despite the small number of team-level observations. This finding suggests that embedding Socratic questioning into LLM-driven agents can enhance the rigor and coherence of students' collaborative research design outputs, even under authentic classroom constraints. However, the small sample size in this study (n=8 per group) limits statistical power and may reduce generalizability of the findings. While the observed effects are statistically significant, caution is needed when interpreting these results. Future research with larger and more diverse samples is necessary to confirm the robustness and generalizability of the findings.

5.2. Results on RQ2

5.2.1. Reflective thinking assessment

To address RQ2, we compared EG and CG in post-test reflective thinking scores. The Shapiro–Wilk tests confirmed that all four reflective thinking subscales were normally distributed in both EG ($n = 48$) and CG ($n = 46$; all $p > 0.05$). Levene's tests indicated no significant variance inequalities across subscales (max $F = 1.017$, $p = 0.318$), and tests of homogeneity of regression slopes showed no significant interactions (max $F = 1.108$, $p = 0.297$), thereby justifying the use of ANCOVA.

As shown in Table 4, after controlling for pretest scores, the EG's overall reflective-thinking score (unadjusted $M = 4.35$; adjusted $M = 4.359$) was significantly higher than the CG's (unadjusted $M = 3.80$; adjusted $M = 3.819$; $F = 9.218$, $p = 0.003$, $\eta^2 = 0.114$). Subscale analyses revealed modest group differences in Habitual Action ($F = 5.194$, $p = 0.025$, $\eta^2 = 0.056$) and Understanding

Table 4
ANACOVA analysis of scores in reflective thinking.

Variable	Group	N	M	SD	Adjusted mean	SE	F	η^2
Overall Reflective Thinking	Experimental	48	4.35	0.37	4.359	0.091	9.218	0.114
	Control	46	3.80	0.42	3.819	0.089		
Habitual Action	Experimental	48	3.95	0.40	3.954	0.092	5.194	0.056
	Control	46	3.75	0.45	3.769	0.091		
Understanding	Experimental	48	4.20	0.36	4.198	0.089	7.003	0.072
	Control	46	3.95	0.39	3.949	0.088		
Reflection	Experimental	48	4.40	0.38	4.409	0.087	11.438	0.124
	Control	46	3.70	0.43	3.695	0.090		
Critical Reflection	Experimental	48	4.25	0.35	4.238	0.086	13.712	0.137
	Control	46	3.69	0.36	3.698	0.079		

($F = 7.003$, $p = 0.010$, $\eta^2 = 0.072$). More pronounced differences emerged in the higher-order dimensions: Reflection ($F = 11.438$, $p = .001$, $\eta^2 = 0.124$) and Critical Reflection ($F = 13.712$, $p < 0.001$, $\eta^2 = 0.137$).

Because these outcomes were based on self-reported questionnaires, the results should be interpreted as reflecting participants' perceived levels of reflective engagement rather than objectively observed performance. Overall, students in the S-ICA condition reported higher reflective thinking, particularly in the advanced dimensions of Reflection and Critical Reflection, suggesting that the use of Socratic questioning within the ICAs may support deeper self-perceived cognitive engagement.

5.2.2. Reflective thinking process performance

Fig. 9 presents the ENA spatial distributions for the EG and CG as they interacted with their respective ICAs, illustrating differences in the reflective thinking structures exhibited during dialogue. For ease of comparison, each EG student is represented by a red dot and each CG student by a blue dot, corresponding to their individual cognitive-network projection in ENA space. The centroid of the EG's average cognitive network is indicated by a solid red square, while the CG's centroid is shown as a solid blue square. These two centroids are clearly separated in space; the dashed rectangles surrounding each square denote the 95% confidence interval for each group, further underscoring group differences in cognitive strategy and reflective language use. A two-sample t-test confirmed this divergence. Along the ENA X-axis (MR1), the EG's mean network score (M = 0.36, SD = 0.25, N = 48) was significantly higher than the CG's mean (M = –0.39, SD = 0.30, N = 46), $t(92) = 13.08$, $p < 0.001$, Cohen's $d = 2.75$. This large effect size indicates that the EG significantly outperformed the CG in terms of cognitive co-occurrence structure. Along the ENA Y-axis (SVD2), a two-sample t-test assuming unequal variances showed no statistically significant difference between the EG (M = 0.00, SD = 0.37, N = 48) and the CG (M = 0.00, SD = 0.24, N = 46), $t(81.29) = 0.00$, $p = 1.00$, Cohen's $d = 0.00$.

To more effectively explain the cognitive differences between the groups in the ENA network structure and identify how the EG and CG differed in their reflective thinking pathways when using different ICAs, we conducted a differential network analysis. By subtracting the CG's co-occurrence network from the EG's (EG minus CG), we generated a contrast network where red edges denote node-pair connections stronger in the EG's network, while blue edges indicate stronger connections in the CG's network. The thickness of the edges corresponds to the magnitude of the difference and co-occurrence frequency.

As shown in Fig. 10(a), the ENA network analysis reveals distinct differences in the cognitive pathways activated by the S-ICA and nS-ICA groups. In the EG, red edges represent stronger connections between key reflective thinking categories, indicating a more robust and integrated cognitive structure. As detailed in Table 5, we further examined and compared the connectivity patterns and strengths for each node pair across both groups' average network graphs. Fig. 10(b) illustrates that, for most node pairs, the EG exhibited higher mean connection strength than the CG, highlighting greater co-occurrence density and cognitive integration during reflective dialogue. Specifically, the EG showed tighter interconnections among multiple levels of reflective thinking—particularly between Understanding and Reflection (U–R, 0.53), Reflection and Critical Reflection (R–CR, 0.43), and Understanding and Critical Reflection (U–CR, 0.32). These connections suggest that EG students, supported by the S-ICA, more frequently activated mid- to higher-order reflective cognitive pathways.

In contrast, the CG displayed comparatively stronger connections in low-order pathways, particularly between Habitual Action and Understanding (HA–U, 0.65) and between Habitual Action and Reflection (HA–R, 0.35), as shown in Fig. 10(c). These findings reveal that, while the CG activated more basic cognitive processes, the S-ICA group was able to develop more sophisticated reflective thinking pathways, reinforcing the effectiveness of Socratic prompting in fostering higher-order cognitive engagement.

These findings support the hypothesis that Socratic prompting enhances cognitive engagement by encouraging students to explore and critically reflect on their understanding. The ENA analysis offers a robust framework for identifying and interpreting the cognitive pathways activated by the S-ICA, revealing the underlying cognitive mechanisms that facilitate more sophisticated levels of reflective thinking.

5.3. Results on RQ3

An analysis of covariance (ANCOVA) was conducted on posttest learning motivation scores, using pretest motivation as a covariate. Prior to running ANCOVA, Levene's test for homogeneity of variances revealed no significant difference between the groups (Levene's F = 0.728, $p = 0.396 > 0.05$), satisfying the equal-variance assumption. The test for homogeneity of regression

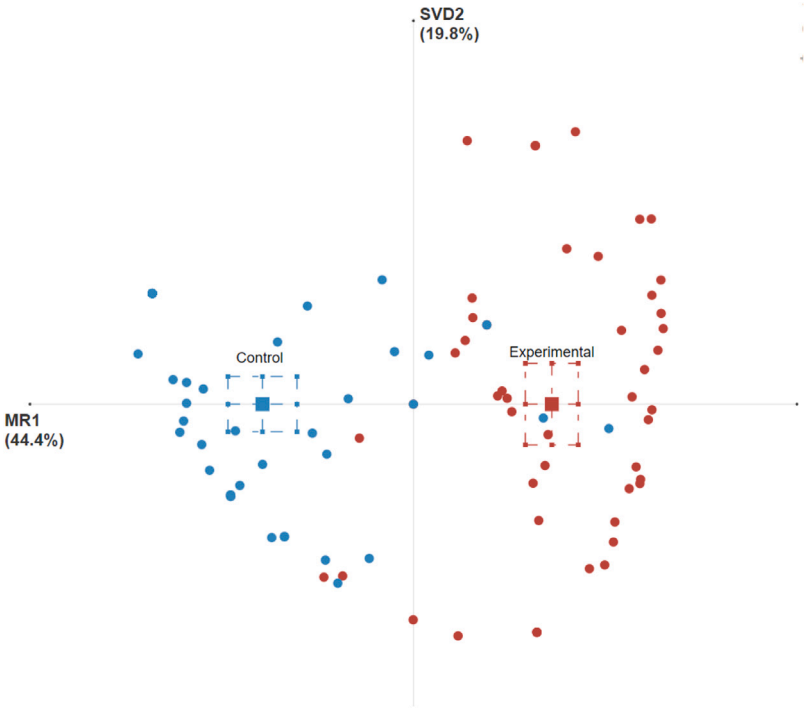


Fig. 9. Mean of the plotted points for the EG and CG.

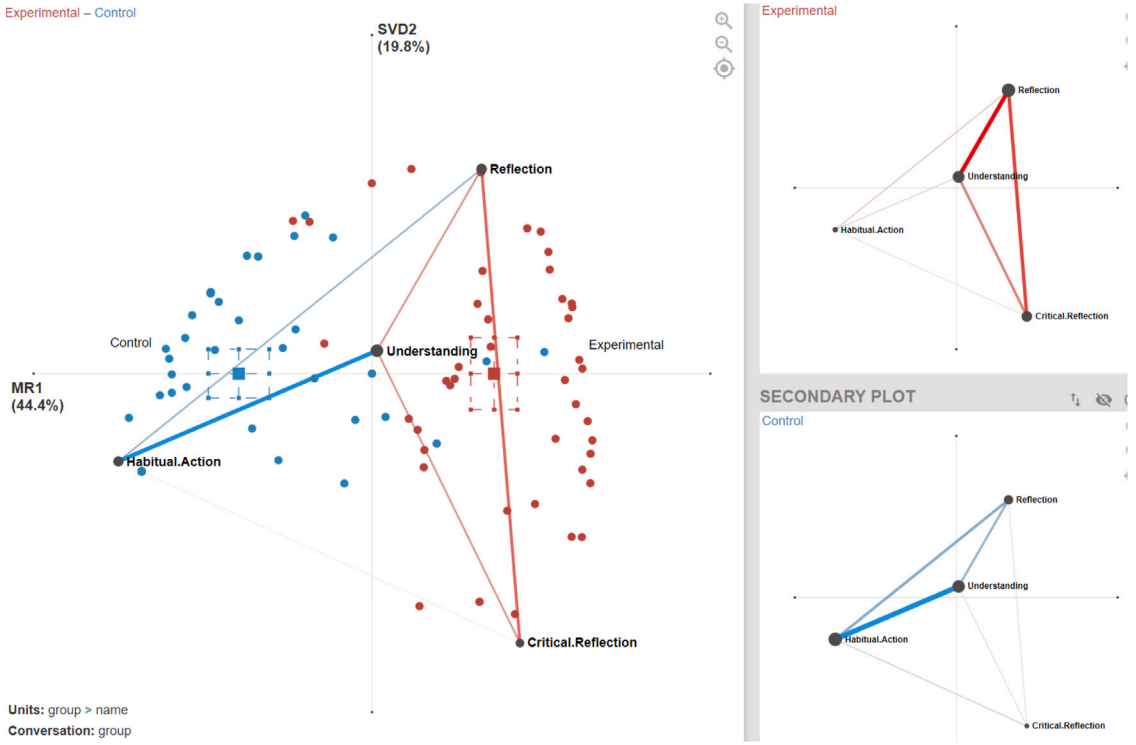


Fig. 10. ENA networks of the EG and CG. (a) The comparison networks of the EG and CG, (b) The mean network of the EG, (c) The mean network of the CG.

Table 5
Connection coefficients of ENA networks.

Connection	Experimental group (S-ICA approach)	Control group (nS-ICA approach)
HA-U	0.11	0.65
HA-R	0.13	0.35
HA-CR	0.08	0.11
U-R	0.53	0.29
U-CR	0.32	0.08
R-CR	0.43	0.07

Table 6
ANCOVA results for learning motivation scores.

Group	N	M	SD	Adjusted M	SE	F	*p*-value	η^2
Experimental (S-ICA)	48	4.01	0.42	3.924	0.096	1.749	0.190	0.041
Control (nS-ICA)	46	3.77	0.49	3.818	0.093			

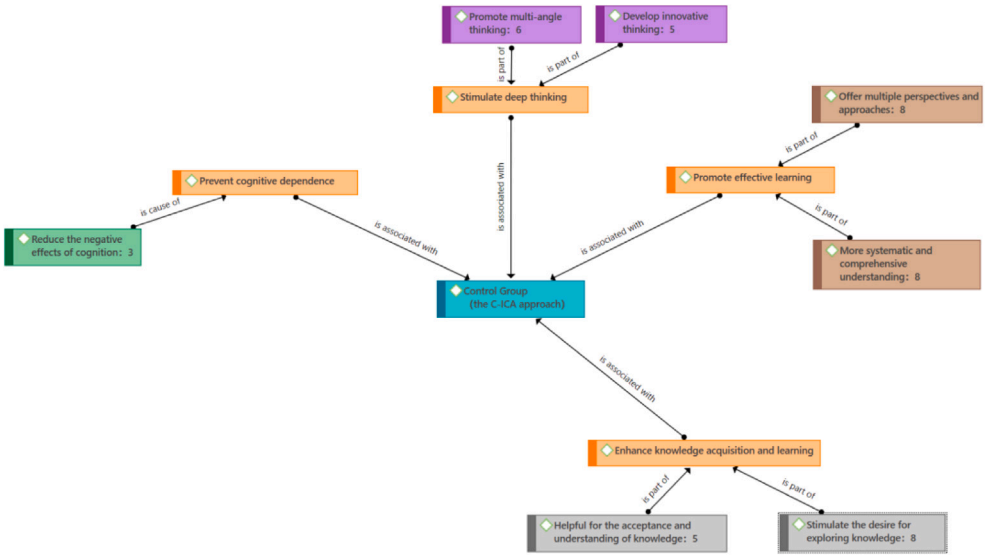


Fig. 11. The control group interview results.

slopes also indicated no significant interaction between the covariate and group (F = 0.603, $p = 0.440 > 0.05$), validating the use of ANCOVA.

As reported in Table 6, the ANCOVA results showed no statistically significant difference in adjusted mean motivation scores between the EG ($M_{adj} = 3.924$, SE = 0.096) and the CG ($M_{adj} = 3.818$, SE = 0.093; F = 1.749, $p = 0.190 > 0.05$). In other words, participants in the S-ICA condition did not report higher levels of learning motivation compared with those in the nS-ICA condition. This suggests that while the S-ICA may enhance cognitive outcomes such as knowledge and reflective thinking, its influence on learners' motivational perceptions appears limited within the scope of this study.

5.4. Results on RQ4

We conducted content analysis on the transcribed interview data using ATLAS.ti 25 for inductive qualitative analysis. First, open coding was employed to group codes with similar meanings into themes and subthemes. After excluding irrelevant codes, the EG interviews yielded five themes and eleven subthemes, while the CG interviews yielded four themes and seven subthemes. We used ATLAS.ti 25 network feature to visualize axial coding, linking coded themes (blue nodes) with their respective subthemes (orange nodes). Fig. 11 and 12 depict the interview results for both groups. Finally, we compared the two groups' reported learning experiences and perceptions, distilling the following five overarching themes.

5.4.1. Stimulate deep thinking

During discipline-specific coursework, using the S-ICA to support learning more effectively stimulates students' deep thinking. Its persistent, probing questions and counter-questions not only encourage students to generate additional ideas in response to the dialogue's trajectory but also progressively prompt them to explore and expand their thinking patterns. For example, one participant



Fig. 12. The experimental group interview results.

(S120506) reported that the agent's questions guided him to re-examine his experimental design step by step; as the agent's inquiries deepened, the critical issues in his design became increasingly evident, enabling him to make targeted corrections. Another student (S120510) noted that using the S-ICA "fostered a desire to engage in dialogue and to answer questions", which in turn broadened his line of thinking and sparked creative insights. A third participant commented, "The agent's continuous questioning gradually broke my habitual thinking patterns; when I faced complex problems, I began to analyze them from multiple angles, truly upgrading my cognitive processes".

5.4.2. Promote effective learning

In terms of promoting effective learning, a comparison of interview data and coding frequencies between the two groups revealed that only students who used the S-ICA experienced genuinely effective learning. For instance, S120403 stated that "digging into a concept by asking 'why' at every step" helped him develop a deeper understanding of that concept. S120402 similarly reported that "interactive dialogue enabled me to internalize the material more profoundly", facilitating knowledge acquisition. Conversely, in the control group (using the nS-ICA), some students acknowledged that the agent significantly expedited information access and improved their ability to retrieve resources; however, they also remarked that such superficial learning "did not truly occur" (C120405). Another CG student (C120501) observed that "using a generic conversational agent resulted in surface-level learning without generating more creative solutions, rendering the process effectively ineffective".

5.4.3. Raise doubts and reflections

Throughout the learning process with the S-ICA, experimental-group students were more inclined to generate questions and engage in reflective thinking. The agent's Socratic counter-questions enabled students to continuously revisit and re-evaluate their initial ideas, prompting them to scrutinize the underlying rationale and intentions. One student (S120503) remarked, "As I interacted with the agent, it kept questioning each element of my experimental design. In responding to these queries, I began to doubt my original logic and thought carefully about how I had planned my study". Another student (S120508) noted, "While I was clarifying my experimental reasoning with the S-ICA, I was constantly reflecting on the logic and shortcomings of my thought process. This caused me to question and delve deeply into the conceptual relationships behind each idea, greatly expanding my cognitive horizons".

5.4.4. Enhance knowledge acquisition and learning

Compared with those using the nS-ICA, students in the experimental group demonstrated stronger knowledge internalization and cognitive construction. Most S-ICA users reported that they could better master the target knowledge and enhance their understanding of the content. For example, one student reported, "During my learning, I continually unearthed and verified additional experimental factors. By actively searching for information and discussing with teammates, I built a more comprehensive, systematic theoretical framework" (S120507). In contrast, a control-group student (C120406) stated, "The nS-ICA's guidance was simplistic and mechanistic; although it offered slightly more detailed direction, I ended up completing the task without actually learning anything". Another CG participant noted, "In our subgroup, some members simply copied the agent's generated answers, which further hindered genuine knowledge mastery" (C120405).

5.4.5. Prevent cognitive dependence

During the coding of interview data, students in both groups frequently mentioned "cognitive dependence". EG students felt that using the S-ICA effectively mitigated the cognitive dependence and degradation associated with general large-language-model tools (frequency of occurrence among EG interviews: 18). For example, S120403 reported, "With a generic agent, it's easy to fall into a dependence on the tool and gradually lose independent thinking ability. But using the S-ICA immerses me in a highly authentic dialogue—so I cannot simply obtain the final answer, and my cognitive skills are exercised instead". Overall, nearly every student in the experimental group remarked that "the system prevented me from relying on ready-made answers". CG participants similarly mentioned cognitive dependence (frequency in CG interviews: 15), observing that reliance on a generic conversational agent stifled innovation and independent thought.

6. Discussion

6.1. Academic performance

The findings indicate a differentiated impact of the S-ICA across outcome domains. At the individual level, both groups achieved substantial gains in conceptual knowledge, but the advantage of the S-ICA over the nS-ICA was modest and only marginally detectable. This pattern is consistent with prior research showing that GenAI-based tools can enhance learner achievement, yet their effects on conventional knowledge tests are typically small to moderate (Hwang & Chen, 2023). Such tests tend to privilege the correctness of final answers rather than the justificatory and regulatory processes that Socratic dialogue is designed to cultivate (Lin et al., 2023). As a result, the improvements in epistemic regulation and reasoning depth elicited by the S-ICA may not have been fully captured by this form of assessment.

Taken together, the findings indicate that the educational benefits of Socratic scaffolding manifest differently across outcome levels. While conceptual knowledge tests revealed only modest incremental gains for the S-ICA, the group-authored proposals demonstrated a substantial advantage, underscoring the value of dialogic scaffolding in tasks that demand extended reasoning, integration of perspectives, and collaborative knowledge construction. Pedagogically, this points to a productive sequencing strategy: employing S-ICA to foster individual reflection and epistemic accountability prior to group collaboration, so that collective work can build on stronger, more critically reasoned contributions. More broadly, the results illustrate how GenAI-driven conversational agents, when designed with adaptive questioning strategies, can personalize the level of cognitive challenge and support learners in ways that accumulate into higher-quality collective outcomes Hu & Shao, 2025. This dual role—providing individualized scaffolding while enhancing group performance—resonates with established traditions in scaffolding and collaborative learning (Molenaar et al., 2014) and highlights a promising direction for leveraging GenAI to advance both personalized and collaborative dimensions of academic performance. Although this study provides valuable insights into the impact of Socratic questioning in LLM-driven agents, the small sample size necessitates caution in drawing broad conclusions. Future studies with larger sample sizes are needed to validate these results.

6.2. Reflective thinking

Results indicated that students supported by the S-ICA reported significantly higher levels of reflective thinking than those in the control group, with the most pronounced differences emerging in the advanced dimensions of Reflection and Critical Reflection. This suggests that embedding Socratic dialogue strategies into an ICA can foster deeper forms of self-reported reflective engagement. Such an interpretation is consistent with prior studies showing that sustained questioning techniques stimulate learners' critical reasoning and self-reflective capacities (Hu, 2023). Similarly, Rospigliosi (2023) demonstrated that LLM-based agents such as ChatGPT can promote iterative cycles of "question–answer–reflection", thereby supporting students' awareness of their own thinking. The theoretical underpinnings of Socratic pedagogy also reinforce these findings: as Edmunds (2006) emphasized, the Socratic method positions the teacher—or in this case, the agent—as a facilitator who provokes cognitive conflict and challenges prior assumptions. Through this process, learners are encouraged not only to articulate what they already know but also to engage in higher-order conceptual inquiry and metacognitive regulation. Other scholars have similarly highlighted how dialogic approaches promote consensus-building and cognitive engagement through dialectical reasoning, social negotiation, and self-correction (Darmawansah et al., 2024).

Beyond theoretical support, our ENA analyses provided empirical evidence for these patterns. The experimental group exhibited stronger network connections among key pathways—notably Understanding–Reflection, Reflection–Critical Reflection, and Understanding–Critical Reflection—compared with the control group. These findings suggest that the S-ICA more effectively activated mid- to higher-order reflective pathways. This observation accords with (Moon et al., 2000)'s model of reflective learning, which characterizes reflection as an iterative and nonlinear process of reconstructing meaning. As Seeskin (1987) noted, Socratic dialogue "thaws frozen thought", encouraging learners to pause, reconsider, and deepen their reflection. Accordingly, the present study contributes to calls for integrating Socratic questioning and other philosophical frameworks into AI-mediated instruction to cultivate reflective thinking.

Further examination of the ENA network differences reveals the underlying cognitive mechanisms at play. In the control condition, where students interacted with a nS-ICA that provided direct feedback without any prompting for reflection, the network exhibited fewer interconnections and weaker reflective pathways. This observation underscores the critical role of Socratic questioning in stimulating deeper cognitive engagement. The S-ICA, by contrast, encouraged a more complex, recursive interaction

between students' initial understandings, reflective thought, and critical evaluation. The iterative process of query–response–revision observed in the experimental group represents the core pedagogical mechanism that enhanced reflective thinking. This process mirrors concept of reflective practice, where students actively engage in self-regulation and self-correction to refine their reasoning (Schön, 1995).

These results suggest that the observed improvements in reflective thinking and cognitive integration were not merely the product of LLM technology but arose from the integration of Socratic scaffolding with LLM-based conversational agents. The S-ICA does not simply serve as a passive content-delivery tool but actively shapes the learning process by fostering higher-order thinking. This interaction between LLM technology and Socratic questioning offers a compelling model for AI-mediated instruction that supports critical thinking and reflective learning. Besides, the ability of Socratic prompting to activate mid- to higher-order thinking pathways is pivotal in deepening students' understanding and critical thinking skills, which are essential for academic success in complex learning environments.

6.3. Learning motivation

The present study found no statistically significant difference in reported learning motivation between students using the S-ICA and those using the nS-ICA. This contrasts with previous research suggesting that intelligent conversational agents and GenAI tools can, under certain conditions, enhance motivation and related affective outcomes such as self-efficacy and learning attitudes (Essel et al., 2022). In our context, however, the motivational impact of the S-ICA appeared limited, despite its positive effects on cognitive outcomes.

One plausible explanation lies in Self-Determination Theory, which emphasizes that sustained intrinsic motivation requires the fulfillment of autonomy, competence, and relatedness needs (Deci & Ryan, 2012). While the S-ICA's progressive questioning may have promoted cognitive engagement, it may not have adequately supported students' sense of autonomy or personal relevance. In other words, dialogic interaction with an agent, in the absence of stronger alignment with learners' interests or values, may be insufficient to trigger meaningful increases in motivation (Vansteenkiste et al., 2004). This limitation may have been amplified by the nature of the course context—"educational research methods"—which many students perceive as a technical requirement rather than an intrinsically appealing subject domain Hsieh et al. (2012).

Taken together, these results suggest a differentiated impact of the S-ICA: while effective in enhancing conceptual knowledge and the quality of collaborative outputs, its influence on students' motivational perceptions remained negligible. This finding highlights the importance of designing GenAI-supported learning environments that combine cognitive scaffolding with targeted motivational supports, for example, by incorporating greater personalization, adaptive goal-setting, or affective feedback mechanisms. Future work should explore whether longer interventions or stronger personalization features enable S-ICAs to foster not only cognitive development but also more robust motivational engagement.

6.4. Learning perceptions

To complement the quantitative findings, semi-structured interviews were conducted to capture students' perceptions of their learning experiences. Participants in the experimental group emphasized that the S-ICA, through sustained questioning, refutation, and heuristic prompts, encouraged deeper engagement during proposal evaluation. They reported that the system facilitated internalization of research concepts, clarification of variable relationships, and more active participation in knowledge construction. In contrast, students in the control group described the nS-ICA as useful for task completion but noted tendencies toward "information dependence" and "cognitive passivity". Several admitted to directly adopting AI-generated responses without critically engaging with the underlying research logic, which was reflected in weaker rigor and feasibility of their proposals. These accounts align with prior work suggesting that while AI can expedite access to information, effective learning requires structured guidance to transform information into deeper understanding (Lin, 2019).

Moreover, interviews revealed potential affective and motivational benefits of the S-ICA. Many students noted that, although continuous questioning initially imposed cognitive load, it ultimately "forced me to think carefully", thereby heightening their engagement with the task (Belland et al., 2013). This "cognitive challenge + guidance" model aligns with (Delen et al., 2014) description of immersive learning environments, where strong interactivity and guided scaffolding alleviate feelings of isolation and boost motivation. While the quantitative results did not show a significant overall increase in learning motivation, the qualitative accounts revealed that some students nevertheless experienced heightened interest, persistence, and exploratory willingness when interacting with the S-ICA. This divergence highlights the distinction between measurable motivational outcomes and students' subjective experiences. It also underscores the need, when optimizing intelligent conversational agents, to balance cognitive rigor with user-friendly interactions, thereby enhancing system acceptability and sustainable use. Future research should further investigate how adaptive learning mechanisms—such as tailoring questioning frequency, depth, and feedback to individual learner profiles—can translate these localized perceptions of increased engagement into sustained and generalizable motivational gains.

6.5. Research implications

The findings of this study suggest that the integration of LLM-based intelligent conversational agents with Socratic questioning strategies holds potential for enriching higher education learning processes, though the benefits appear differentiated across outcome domains. At the individual level, conceptual knowledge gains were substantial across both conditions, with only a modest and at best marginal advantage for the S-ICA. At the collective level, however, teams supported by the S-ICA produced significantly higher-quality research proposals, indicating that dialogic scaffolding may be particularly effective for open-ended, collaborative tasks requiring justification and integration of ideas. In contrast, no statistically significant improvements were observed in students' self-reported learning motivation, suggesting that the cognitive benefits of Socratic prompting did not automatically extend to affective dimensions.

Theoretically, these findings contribute to ongoing debates on how GenAI can be meaningfully combined with established pedagogical traditions. By empirically examining the role of Socratic strategies in an LLM-mediated environment, this study illustrates how dialogic scaffolding can strengthen epistemic processes such as justification and reflection, which may subsequently accumulate into more coherent collective outputs. At the same time, the limited motivational impact observed underscores the need to more explicitly align AI-supported learning designs with students' autonomy, competence, and relatedness needs, as emphasized in motivational theory. These nuanced outcomes highlight that GenAI's educational value is not uniform but depends on the specific learning processes targeted and the assessment methods employed.

Practically, the results yield implications for integrating AI-driven agents into university instruction. Replication of this approach requires not only technical implementation of prompt-engineering strategies but also contextual embedding in authentic coursework, ensuring alignment with curricular goals and student needs. Rather than being constrained by the study's scope—a six-week intervention and discipline-specific task—the findings point to promising avenues for extending such designs to broader contexts. Future implementations should investigate how stronger personalization, adaptive feedback, and longer exposure periods affect both cognitive and motivational outcomes, as well as whether the observed team-level benefits transfer to other forms of academic work. In particular, adaptive learning systems could play a pivotal role in tailoring feedback to learners' prior knowledge, interests, and progress trajectories, thereby enhancing both performance and motivation. More broadly, the study underscores the importance of designing GenAI-supported environments that balance immediate cognitive scaffolding with mechanisms for sustaining motivation and promoting transferability across domains.

Overall, the present study highlights both the promise and the boundaries of embedding Socratic strategies into LLM-based agents. While the approach demonstrated clear potential to enhance cognitive and collaborative outcomes, its effectiveness will depend on careful contextualization, alignment with motivational principles, and sustained exposure. Future work should therefore position such integrations not as universally transformative solutions, but as targeted pedagogical innovations whose value lies in their capacity to enrich personalized and adaptive learning when thoughtfully implemented.

7. Limitations and future directions

Several limitations should be acknowledged. First, the sample was primarily composed of students majoring in educational technology, and the gender distribution across participants was uneven. This sampling context may limit the generalizability of the findings to learners from other disciplines and backgrounds. Future research should therefore aim to recruit more diverse and representative samples to enhance external validity.

Second, the instructional tasks implemented in this study predominantly focused on research design-oriented activities. Although this context aligns well with the development of reflective thinking and higher-order reasoning, it remains uncertain whether comparable benefits would be observed in other domains of learning, such as discipline-specific content knowledge, professional skills, or affective outcomes. Future studies should therefore extend the application of S-ICAs to a wider range of instructional settings to examine transferability and ecological validity.

Third, the relatively short duration of the intervention limited the ability to examine the long-term effects of the AI-supported Socratic dialogue agent on learners' reflective practices and sustained behavioral change. Follow-up studies should adopt longitudinal designs and repeated measures to capture whether the observed gains in reasoning and proposal quality persist, attenuate, or compound over time.

Fourth, the limited number of group-level submissions (eight proposals per condition) constrained statistical power and reduced the precision of effect estimates. While robust analytic techniques were employed, future research should increase the number of groups to better account for group-level variance and enhance the reliability of inferences.

In addition, ethical considerations surrounding LLM-based learning processes were not comprehensively addressed in this research. Issues such as data privacy, content originality, and intellectual property rights pose potential challenges to the responsible use of GenAI-generated content in educational contexts. Accordingly, future work should not only investigate how learners and educators perceive these ethical risks, but also develop practical guidelines and institutional policies to ensure that GenAI tools are adopted responsibly, transparently, and in ways that safeguard academic integrity.

Table 7
Evaluation rubric for experimental research proposals.

Criteria	Excellent*(Score = 10)*	Good *(Score = 8)*	Satisfactory*(Score = 6)*	Needs Improvement*(Score = 4)*	Poor*(Score = 2)*
Research Topic	The title is very concise, clear, and standard; accurately reflects the experimental scope and significance.	The title is concise, clear, and appropriate; reflects the experimental scope and significance.	The title is moderately clear; reflects part of the experimental scope.	The title lacks clarity or completeness; scope is vague.	The title is vague or overly complex; does not reflect the scope.
Research Question	Clearly and precisely defines the experimental research question; variables and hypotheses are explicitly stated.	Precisely defines the research question; main variables and hypotheses are identified.	The research question is generally clear; variables partially identified.	The research question lacks clarity; variables or hypotheses are insufficiently stated.	The research question is vague; lacks variables and hypotheses.
Theoretical Rationale	The literature comprehensively reviews key concepts and variables; strongly supports the experimental design.	The literature sufficiently reviews relevant concepts and variables; supports the design.	The literature partially reviews relevant concepts; limited support for the design.	The literature is insufficient or lacks focus on key variables.	The literature is outdated or irrelevant; no conceptual basis provided.
Research Content	Experimental variables and design are fully specified; selection is highly scientific and appropriate.	Variables and design are clearly specified; selection is appropriate.	Variables and design are generally specified; selection is partially appropriate.	Variables and design are unclear or incomplete; selection is questionable.	Variables and design are vague or unsuitable.
Research Methodology	Methods are rigorously selected and described; clearly aligned with experimental goals; appropriate model is detailed.	Methods are appropriate and well described; the model is appropriate.	Methods are generally appropriate; the model is described briefly.	Methods are insufficiently described or only partially appropriate; the model lacks clarity.	Methods are vague or inappropriate; the model is missing.
Research Procedures	Procedures are comprehensive, detailed, and feasible; clear implementation plan and preparation steps are provided.	Procedures are clear and feasible; implementation steps are adequate.	Procedures are moderately clear; some steps are outlined.	Procedures lack detail or clarity; feasibility is limited.	Procedures are vague or not implementable.

8. Conclusion

Overall, the primary contribution of the present study lies in the design and empirical validation of the S-ICA that integrates Socratic dialogue strategies with LLM technology to enhance students' research design skills, reflective thinking, and perceptions of learning tools within a discipline-specific course. Results indicate that the S-ICA confers significant advantages in guiding students through higher-order cognitive tasks, stimulating deep reflection, and improving the scientific rigor of research proposals—particularly along the dimensions of Reflection and Critical Reflection.

A distinctive feature of the S-ICA is its capacity to provide personalized and adaptive scaffolding: by semantically parsing learner input and tailoring subsequent questions to individual reasoning stages, the system was able to align instructional support with students' needs, strengths, and levels of preparedness. This adaptive responsiveness ensured that feedback was neither overly simplistic nor excessively demanding, thereby sustaining engagement and promoting productive struggle. In this sense, the study contributes to the growing body of work on AI-supported personalized learning by demonstrating how dialogic scaffolding can adapt in real time to heterogeneous learners.

In the context of the rapid advances in GenAI, a central challenge for educational research is how to leverage LLM-based conversational systems to enhance learning without fostering cognitive dependency or encouraging unethical academic practices. This study offers a model for integrating classical instructional strategies with cutting-edge AI technologies, providing empirical evidence of their potential efficacy in educational settings. The Epistemic Network Analysis (ENA) results demonstrated that Socratic prompting effectively activated more advanced reflective pathways, indicating that such AI-driven systems can play a crucial role in higher education by fostering critical thinking and reflective engagement. Furthermore, the findings suggest that adaptive Socratic prompting could strike a balance between two essential goals in higher education: supporting personalized learning paths for individual students while simultaneously promoting collective knowledge construction in collaborative tasks.

This approach addresses concerns about the potential risks of AI-generated content in education (Abbas et al., 2024; Ng et al., 2024) by demonstrating how structured, question-driven dialogue can mitigate those risks while promoting deeper learning. Future research should extend this line of inquiry by examining how adaptive Socratic ICAs can be scaled to diverse subject areas, longer interventions, and varied learner populations, thereby advancing the integration of personalized and ethical AI practices in higher education.

CRediT authorship contribution statement

Linjin Xi: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation, Conceptualization. **Yi Zhang:** Supervision, Resources, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization. **Qiyun Wang:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

Ethic statement

The research involved human participants and followed the procedures of the Education Faculty's Ethics Committee. Informed consent was obtained for all participants.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

See Table 7.

Data availability

Data will be made available on request.

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